

Agentic AI: A Perspective on Architecture, Frameworks and Applications

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Comparative empirical evaluation of LangChain, LangGraph, and CrewAI measuring orchestration depth, memory reuse, and branching latency.

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

60 screened papers and live market analysis case study.

Key Numerical Findings & Empirical Proofs:

Quantifies architectural trade-offs across vertical, horizontal, and hybrid agent coordination graphs.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Confirms all leading frameworks use hardcoded DAG branching without dynamic epistemic uncertainty gating; excludes AutoGen and DSPy.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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